

Inference as Consolidation: Replay in the Silent Degrees of Freedom

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Abstract

Biological brains consolidate memories both in dedicated offline states (sleep) and, as recent causal evidence shows, through *local sleep*: brief, use-dependent off-periods of individual cortical circuits during wakefulness. In contrast, replay-based continual learning in artificial neural networks almost universally relies on offline consolidation phases or on interleaving replayed samples with the input stream. We ask whether a network trained end-to-end by local, biologically constrained learning rules can consolidate *during inference*, with no offline phase at all. We introduce (i) a *refractory rotation* rule under which units that have just fired sit out the next competition, (ii) an *isolation* scheme that confines replay updates to hidden synapses invisible to the current input under k -winner-take-all (k -WTA) dynamics — exact for silent presynaptic and suppressed units, and measured to alter the waking hidden code of only 0.2–0.4% of samples per replay step elsewhere — with per-synapse optimiser state advanced only inside the mask, and (iii) internal control signals — a unit-level homeostatic pressure that triggers replay bursts, and a relative-novelty (fast-over-slow error) gate that decides when rotation runs. The construction inverts the usual direction of non-interfering continual learning: rather than protecting past tasks while learning the present one, the present computation is held invariant while past memories are consolidated through degrees of freedom the current batch exposes as unused. On class-incremental split-MNIST the resulting system reaches $92.7 \pm 0.3\%$ (six seeds) with no offline phase, exceeding the best offline-night schedule ($90.7 \pm 0.4\%$) at a replay-sample cost of $\approx 1.2\times$ the night’s; reversing the protected direction (guarding past memories while learning the present, as in null-space methods) costs fifteen points under identical machinery; masked replay is insensitive to replay batch size down to eight samples, whereas unmasked replay and offline nights degrade sharply at the same sizes. Rotation’s i.i.d. price is governed by the substrate’s activity fraction: at the 5% fraction the project inherited, novelty gates (masked Adam) or sleep-out-committed gates (stateless SGD) were needed to refund it, whereas at the 10% optimum the price shrinks to about a point and both gates become neutral — the sparsity dial, not the gate, is the cure. On split CIFAR-10 the ordering of methods transfers and widens, and a substrate decomposition shows that the remaining gap to backpropagation-with-replay is dense activation, not credit assignment. Finally, a drive-referenced controller for the rule’s synaptic decay — one dimensionless constant in place of per-dataset tuning — trades at most 1.4 sequential points for the best static accuracies we measure and lifts the system onto 100-class streams that defeat every fixed decay; with locally learned ResNet-style skip synapses the five-hidden-layer stack then matches the two-hidden-layer record on both axes. We provide exactness proofs for the isolation rule and release all code, checkpoints and diagnostics.

1 Problem Formulation

Setting. Let a stream of labelled examples $(x_t, y_t) \in \mathbb{R}^d \times \{1, \dots, C\}$ be presented in T contiguous tasks $\mathcal{T}_1, \dots, \mathcal{T}_T$, where task $\mathcal{T}_\tau$ contains only classes $\mathcal{C}_\tau \subset \{1, \dots, C\}$, the $\mathcal{C}_\tau$ are disjoint, and no

task identity is available at test time (class-incremental learning). A network f_θ with a single shared C -way readout is trained on the stream and evaluated on all C classes after the final task,

$$A_{\text{seq}}(\theta) = \Pr_{(x,y) \sim \mathcal{D}_{\text{test}}} \left[\arg \max_c f_\theta(x)_c = y \right], \quad F = \frac{1}{T-1} \sum_{\tau=1}^{T-1} (a_{\tau,\tau} - a_{T,\tau}), \quad (1)$$

where $a_{t,\tau}$ is accuracy on task τ after learning task t and F is forgetting.

Constraints. We restrict the learner to a set $\mathcal{B}$ of cortical constraints: no weight transport (feedback synapses are learned, never copied), errors carried by a bounded burst-like channel, Dale’s law, sparse population codes (k -WTA on wide layers), sparse distance-dependent connectivity, and a single forward–feedback sweep per example (no iterative relaxation). An episodic memory (“hippocampus”) of at most K raw examples with reservoir writing is permitted; K is a budget, not a free parameter.

Objective. Unlike benchmark-chasing continual learning, we optimise a three-way trade-off

$$\max_{\theta} (A_{\text{seq}}, A_{\text{iid}}, -R, -E) \quad \text{subject to} \quad \theta \in \mathcal{B}, \quad (2)$$

where A_{iid} is accuracy of the identical learner on the i.i.d. shuffle of the same data (the *static axis*: a brain-like mechanism must not damage ordinary learning), R is the replay cost in *samples* $\times$ *synapses*, and E an event-driven energy proxy (Section 3). The central question of this paper: can the offline consolidation term be removed entirely — consolidation occurring only *during* inference on the wake stream — without losing A_{seq} , and at what cost in A_{iid} and R ?

2 Related Work

Sleep-inspired consolidation in ANNs. The Bazhenov line implements sleep as a global offline phase: networks are converted to spiking dynamics and driven by noise under local Hebbian plasticity, recovering old tasks after each new one (Tadros et al., 2022), after several tasks at once (Bazhenov et al., 2026), in spiking networks with coarse alternation of training and sleep (Golden et al., 2022), and in equilibrium-propagation networks combined with small rehearsal buffers (Kubo et al., 2025). Wake–sleep frameworks with NREM and REM stages (Sorrenti et al., 2024; Robinson et al., 2022) and engineering systems such as SIESTA (Harun et al., 2023) likewise schedule consolidation offline. In all of these, sleep is global, offline, and clocked; none implements consolidation confined to currently unused units during ongoing inference.

Gating, subnetwork and history-dependent competition methods. Context-dependent gating (XdG) activates a fixed random subnetwork per task but requires task identity at train and test time (Masse et al., 2018); learned gates (Tilley et al., 2023), dendritic context vectors from input prototypes (Iyer et al., 2022), task-free sparsification by feedforward and top-down errors (Lässig et al., 2023), and gradient-recovered functional masks (McKee et al., 2026) remove the explicit label but derive the gate from the *input identity*. Use-history-dependent suppression of competition, in itself, has classical precedents: refractory winner-take-all dynamics in physiological models of competitive learning (Kaski and Kohonen, 1994), explicit refractory competitive learning in which a recent winner is temporarily barred from winning (Maeda and Miyajima, 1999), and, recently, spiking continual learners whose firing thresholds rise with each unit’s activation history to diversify

recruitment across tasks (Shen et al., 2024). Our rotation differs in *purpose*, not primitive: recently wake-active units are benched for one competition specifically to enlarge a dynamically reallocated *replay channel* — rotation manufactures plastic degrees of freedom in which consolidation can proceed concurrently with inference, rather than allocating representational resources across tasks.

Null-space and parameter-isolation methods. A mature line of work constructs non-interfering updates by protecting *previous* tasks: projecting new-task gradients orthogonally to old-task gradient or activation subspaces (Farajtabar et al., 2020; Saha et al., 2021), training in the null space of accumulated feature covariance (Wang et al., 2021), combining k -winner sparsity and heterogeneous dropout with such projections (Abbasi et al., 2022), or freezing pruned-and-frozen subnetworks (Golkar et al., 2019; Mallya and Lazebnik, 2018). Our construction inverts the protected direction: the *ongoing wake computation* is held invariant while *old-memory replay* is written into degrees of freedom that the current batch’s exact sparse support exposes as instantaneously invisible — no stored bases, SVDs, or task masks, and the “null space” changes every batch for free. McKee et al. (2026) note that optimiser state and decoupled weight decay can break structural guarantees even at zero gradient, and Bricken et al. (2023) analyse stale momentum in sparse learners; we adopt the corresponding requirement (mask-confined moments) as part of the exactness construction rather than claiming its discovery. Replay *scheduling* has likewise been studied as a learned policy (Klasson et al., 2023); our contribution there is the specific unit-level use-pressure controller and its empirical dissociation from novelty triggering.

Complementary learning systems. CLS-ER maintains fast/slow EMA copies of the weights as semantic memories (Arani et al., 2022) and DualNet trains a slow self-supervised learner beside a fast supervised one (Pham et al., 2021); both raise the value of each replayed sample but still replay at every step. Our contribution is orthogonal: we change *where* (isolated synapses), *when* (homeostatic bursts) and at what granularity (micro-batches) replay is applied.

Pattern separation. Cerebellum-, mushroom-body- and dentate-gyrus-style sparse expansions (Cayco-Gajic et al., 2017; Cayco-Gajic and Silver, 2019; Litwin-Kumar et al., 2017) inspire continual learners such as the FlyModel (Shen et al., 2021), sparse distributed memory (Bricken et al., 2023) and DG-gated mixtures (Kapoor et al., 2026). We tested this family as a front-end and report a negative result (Section 5).

Biology of local sleep. Sleep-like off-periods occur locally in awake, sleep-deprived animals (Vyazovskiy et al., 2011). Causal evidence is recent: optogenetically inducing ON/OFF alternation in one cortical hemisphere of awake mice locally discharges sleep pressure, weakens synaptic markers (GluA1, pS845) and rescues memory consolidation under sleep deprivation, whereas an equal *tonic* reduction of firing does not (Driessen et al., 2026). Our refractory rotation is the algorithmic counterpart of this alternation requirement, and our silent-mask control the counterpart of tonic silencing.

3 Experiment Setup

Data and protocol. Sequential axis: split-MNIST (five tasks of two classes, 5 epochs per task) and split CIFAR-10 (same protocol; 32×32 luminance so that distance-dependent connectivity retains its geometry). Static axis: the identical learner and budgets on the i.i.d. shuffle. Waking batches have $n_w = 16$ samples; the episodic buffer holds $K \in \{200, 1000\}$ raw samples under

reservoir writing. All numbers are mean $\pm$ s.d. over three seeds unless marked otherwise; runs are deterministic at a fixed CPU thread count (the reduction order feeds k -WTA’s discontinuity, so a different thread count is effectively a different seed). A third regime uses a frozen V1-like feature front-end for CIFAR-10: 256 normalised random $6 \times 6 \times 3$ patches sampled from the training images act as fixed filters (stride 2, rectification against the per-filter mean response, quadrant average pooling; 1024-D, linear probe $\approx 49\%$). Nothing in the front-end is learned; the cortex under study receives it as its retina. Section 7 additionally uses split CIFAR-100 (ten tasks of ten classes, same protocol and budgets, luminance and feature variants): with tenfold fewer supervision events per class it probes the learning rule far outside the regime its constants were tuned in, and its absolute numbers are floor-regime for every method tested (chance 1%, BP+ER 6–10%) — we use it for orderings and mechanisms, never as a competitive benchmark.

Architecture. d -512-256- C with k -WTA sparsity 10% on the hidden layers — the record fraction identified by the sparsity dial (Section 6.3); the project’s historical 5% fraction survives in the appendices and is marked wherever used — with 30% distance-dependent connectivity on hidden synapses, Dale’s law with 80% excitatory units, and the single-sweep learning rule of Section 4. The offline-night reference uses its own preferred narrower core (256–128).

Baselines. (i) Backpropagation with experience replay at equal buffer K (mandatory in every table); (ii) the offline *night*: after each waking epoch, 20 replay batches of 256 samples at $3 \times$ learning rate with no concurrent input (the strongest schedule from our earlier ablations); (iii) interleaved ER for the local learner; (iv) no buffer.

Metrics. Final accuracy, forgetting F , replay cost R (batches and samples), the isolated-channel width ρ_ℓ (fraction of replay-activated units in layer ℓ that are asleep for the current input), code overlap (mean pairwise Jaccard of task-active unit sets), participation-ratio dimensionality, and an event-driven energy proxy: 0.9 pJ per synaptic event for spiking inference vs. 4.6 pJ per MAC for dense rate inference, with thresholds calibrated post hoc and T_s integration steps (Merolla et al., 2014).

4 Methodology

4.1 Cortical base learner

Layer activities are $a^0 = x$ and, for $\ell = 1, \dots, L$,

$$z^\ell = g^\ell(W^\ell a^{\ell-1}) + b^\ell, \quad a^\ell = \Phi_k(\sigma(z^\ell) \odot (1 - s^\ell)), \quad (3)$$

where σ is a rectifier, $s^\ell \in \{0, 1\}^{n_\ell}$ is the (possibly empty) *suppression mask* of Section 4.3, and Φ_k keeps the k largest entries of each row and zeroes the rest (k -WTA). Errors are produced by one top-down sweep: with target y ,

$$\varepsilon^L = y - z^L, \quad \varepsilon^\ell = \sigma'(z^\ell) \odot \left(B^\ell [\kappa \tanh(\varepsilon^{\ell+1}/\kappa) \odot \mathbf{1}(a^{\ell+1} > 0)] \right), \quad (4)$$

i.e. a bounded, event-gated burst signal carried by learned feedback weights B^ℓ (no transport). Forward and feedback weights follow the same local product with a shared decay λ (Kolen–Pollack alignment (Kolen and Pollack, 1994; Akrouit et al., 2019)),

$$\Delta W^\ell \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda W^\ell, \quad \Delta B^{\ell-1} \propto \varepsilon^\ell (a^{\ell-1})^\top - \lambda B^{\ell-1}, \quad (5)$$

with sign-concordant feedback under Dale’s law and per-synapse adaptive scaling (ϵ -floored Adam). Equations (3)–(5) were fixed by earlier ablation ladders and are not varied here, with one exception: Section 7 replaces the constant λ by an adaptive per-layer controller.

4.2 Exactly isolated replay

Let x be the current waking batch and define the *awake set* $\mathcal{A}^\ell(x) = \{i : a_i^\ell(x) \neq 0 \text{ for some sample in } x\}$. During the same forward step, a replay batch r drawn from the buffer is applied through the synapse mask

$$M_{ij}^\ell = \mathbf{1}[j \notin \mathcal{A}^{\ell-1}(x) \vee i \notin \mathcal{A}^\ell(x)], \quad (6)$$

i.e. a synapse may take the replay update iff its presynaptic unit is silent for the current input or its postsynaptic unit is asleep. We enforce the same constraint on the per-synapse optimiser state (first and second moments advance *only* inside the mask): as McKee et al. (2026) caution and Bricken et al. (2023) analyse for sparse learners, unmasked momentum would otherwise re-inject the isolated update into awake synapses, voiding the exactness guarantees below; the accuracy consequences of this choice are dissected in Section 6. The readout row remains plastic (old classes must stay calibrated).

Proposition 1 (Pre-silent updates are exactly invisible). *Fix input x and let $\widehat{W}^\ell = W^\ell + \Delta^\ell$ with $\Delta_{ij}^\ell = 0$ whenever $a_j^{\ell-1}(x) \neq 0$. Then the network output on x is unchanged for any magnitude of Δ .*

Proof. By induction on ℓ . Assume $a^{\ell-1}$ is unchanged (true for $\ell = 1$ since $a^0 = x$). For any unit i , $\widehat{z}_i^\ell - z_i^\ell = g^\ell \sum_j \Delta_{ij}^\ell a_j^{\ell-1}(x)$. Each term vanishes: if $a_j^{\ell-1}(x) \neq 0$ then $\Delta_{ij}^\ell = 0$; otherwise $a_j^{\ell-1}(x) = 0$. Hence z^ℓ , and therefore a^ℓ through (3), is unchanged; the induction closes at the output layer. $\square$

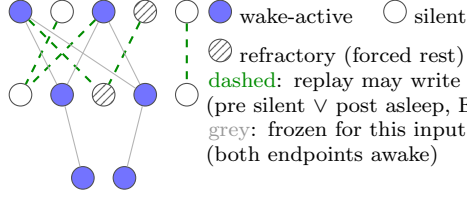
Proposition 2 (Post-asleep updates are invisible under a margin condition). *Let Δ^ℓ additionally allow $\Delta_{ij}^\ell \neq 0$ for units $i \notin \mathcal{A}^\ell(x)$ with active presynaptic partners. Write $\delta_i = g^\ell \sum_j \Delta_{ij}^\ell a_j^{\ell-1}(x)$ and let $z_{(k)}$ be the k -th order statistic of the post-nonlinearity values entering Φ_k . If for every such unit $\sigma(z_i^\ell + \delta_i) < z_{(k)}$ (the perturbed unit stays below the winner threshold), the output on x is unchanged.*

Proof. A unit outside $\mathcal{A}^\ell(x)$ contributes $a_i^\ell(x) = 0$. After the update its pre-activation becomes $z_i^\ell + \delta_i$; under the stated margin it is still excluded by Φ_k (or rectified to zero), so $a_i^\ell(x) = 0$ still, all other units’ inputs are untouched by Proposition 1’s argument, and the layer’s output is identical. $\square$

Corollary 1 (Suppressed units are exact for any magnitude). *If i is suppressed by the rotation mask ($s_i^\ell = 1$ in (3)), then $a_i^\ell(x) = 0$ identically and Proposition 2 holds with no margin condition.*

Remark 1 (How exact, measured). *The default system keeps the readout plastic and does not enforce the margin of Proposition 2, so exactness end to end is an empirical question. On the record configuration, re-inferring every waking batch after each of its 18,760 isolated replay updates (under the suppression it was inferred with): the update altered the top hidden code of 0.36% of the waking samples per step, changed a waking prediction for 0.27% — almost all of it the plastic readout, since the same count is 0.001% with the readout isolated (hidden code 0.21%) — and violated the margin condition for $\sim 10^{-4}$ of the asleep units per step. The isolation is thus exact for silent-presynaptic and suppressed units and near-exact elsewhere; Corollary 1 is why the refractory rotation makes it robust rather than merely approximate. (An earlier logged check that reported zero drift was taken on single batches, where the per-unit flip rate makes zero the typical outcome; the population rates above supersede it.)*

(a) exactly isolated replay



(b) internal control

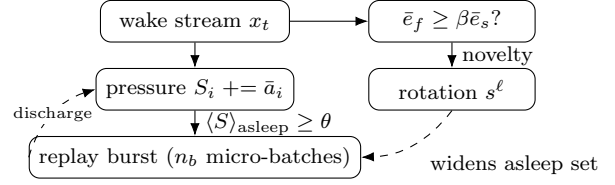


Figure 1: Mechanism. (a) During each waking batch, replay from the episodic buffer is written only into synapses whose update is invisible to the current input — exactly so for silent presynaptic and suppressed units (Propositions 1–2); refractory rotation forces recently used units into the consolidable set (Corollary 1). (b) Two internal signals: unit-level homeostatic pressure triggers replay bursts; a relative-novelty gate decides when rotation runs.

4.3 Refractory rotation and its gates

Definition 1 (Refractory rotation). *After processing waking batch x_t , set $s_i^\ell(t+1) = \mathbf{1}[i \in \mathcal{A}^\ell(x_t)]$: every unit that fired sits out the next competition and is therefore asleep — and consolidable — for batch x_{t+1} .*

Rotation widens the isolated channel ρ_ℓ from 0.2–0.3 (natural silence) to ≈ 0.53 , at the price of running inference on the second-best coalition. Two internal signals control the system (Figure 1):

Homeostatic pressure (when to replay). Each unit integrates its own use, $S_i \leftarrow S_i + \bar{a}_i(x_t)$, and a replay *burst* of n_b consecutive batches is triggered when the mean pressure of currently asleep units exceeds θ ; consolidation discharges the pressure of the units that received it, $S_i \leftarrow (1 - \mathbf{1}[i \text{ asleep}]) S_i$. This is a unit-level Process S (Borbély, 1982).

Relative novelty (when to rotate). With fast and slow EMAs of the batch error $\bar{e}_f(t) = (1 - \alpha_f)\bar{e}_f + \alpha_f e_t$, $\alpha_f = 0.1$, and $\bar{e}_s$, $\alpha_s = 0.002$, rotation runs only while

$$\bar{e}_f(t) \geq \beta \bar{e}_s(t), \quad \beta \approx 1.1\text{--}1.5, \quad (7)$$

i.e. while the stream is *new relative to the learner’s own recent history* (an ACh-like adaptation signal (Hasselmo, 2006)). Monitoring streaming error with exponentially weighted averages is standard in concept-drift detection (Ross et al., 2012), and related loss-ratio triggers appear in concurrent continual-learning work; we claim no priority for the detector itself — only for what it gates (the rotation, hence the replay channel’s width) and for the unmastered clause below. Absolute thresholds provably cannot serve here: the converged error of a ten-class i.i.d. stream exceeds any level a two-class task dips under, and we observe exactly this failure (gate open 100% of the time on static MNIST, 95% on CIFAR, for any threshold that behaves on split-MNIST). Because rotation is *beneficial* on streams the learner never masters (Section 5), the final gate adds an “unmastered” clause: with e_0 the mean error over the first 20 batches (a chance-level estimate), rotation runs while

$$\bar{e}_f(t) \geq \beta \bar{e}_s(t) \quad \text{or} \quad \bar{e}_f(t) \geq \gamma e_0, \quad \gamma \approx 0.5. \quad (8)$$

The first clause opens the gate at task switches, the second keeps it open on any stream whose error remains a substantial fraction of chance — both scale-free.

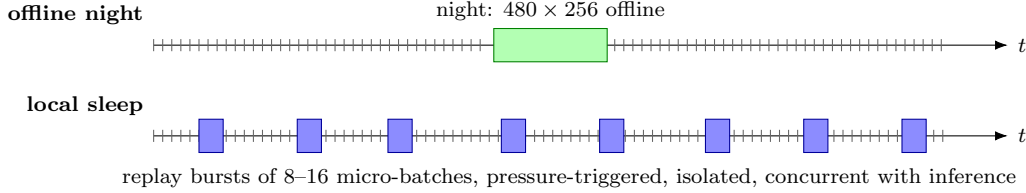


Figure 2: Consolidation schedules. Ticks are waking batches. The offline night pauses the stream; local sleep consolidates during it, in SWR-like bursts confined to asleep units.

4.4 Cost accounting

Replay cost is measured in samples ($R_s = \text{batches} \times \text{batch size}$) and synaptic work. The night costs $R_s = 480 \times 256 \approx 1.2 \times 10^5$ samples. Continuous local sleep with micro-batches of 8 costs $18,760 \times 8 \approx 1.5 \times 10^5$ samples ($1.2\times$ the night); pressure-triggered bursts with batch 64 cost 1.8×10^5 ($1.4\times$).

5 Results

5.1 Local sleep replaces the night

On split-MNIST at $K=1000$, refractory local sleep with an isolated replay micro-batch beside every waking batch reaches $92.7 \pm 0.3\%$ (six seeds) with no offline phase — above the best offline night ($90.7 \pm 0.4\%$ on the night’s preferred narrow substrate; 90.9 ± 3.2 , wildly unstable, on ours) and far above unmasked interleaved ER at the same budget (89.7 ± 1.1). Halving the cadence with doubled replay batches costs 1.3 (91.4 ± 0.3). Adding a night on top of continuous local sleep yields nothing (92.7 ± 0.4 vs. 92.7 ± 0.3): once consolidation runs during wake, the offline phase is redundant at this buffer size. At very small buffers ($K=200$) the night retains an edge only in combination with narrow substrates; the mechanism triangulation is in Appendix C.

5.2 Isolation stabilises micro-batch replay

Figure 3: masked refractory replay is flat in replay batch size from 256 down to 8 samples ($92.3 \pm 0.1 \rightarrow 91.8 \pm 0.3$; it breaks only at 4), while at batch 16 unmasked replay sits at 87.0 ± 1.1 , the silent mask at 88.8 ± 0.8 , and the offline night falls from 90.9 to 87.3 ± 3.2 when its batches shrink to 16. Masking the replay weight update is what makes replay affordable: the earlier “39 $\times$ ” cost of inference-time consolidation was an artefact of counting 256-sample batches.

5.3 The direction of protection matters, and state-free isolation wins

Two ablations sharpen the construction. *Mirror isolation* implements the classical protected direction with our own machinery: the *waking* update is barred from synapses whose both endpoints served the last replay batch (protect the past), while replay itself is applied unmasked. With everything else identical, mirror isolation reaches only $77.9 \pm 1.6\%$ — below even unmasked interleaved ER (89.7 ± 1.1) and far below our direction (92.7 ± 0.3): constraining wake learning impairs the new task while the unmasked replay keeps perturbing the live computation, a double loss. The protected direction is thus not a framing choice but a fifteen-point empirical effect (20 points, with high seed variance 70.3 ± 10.1 , at the historical 5% fraction). (An earlier version of this cell ran with the burst-rate baseline inadvertently frozen during the masked waking update and read 85.0 ± 2.0 ; the

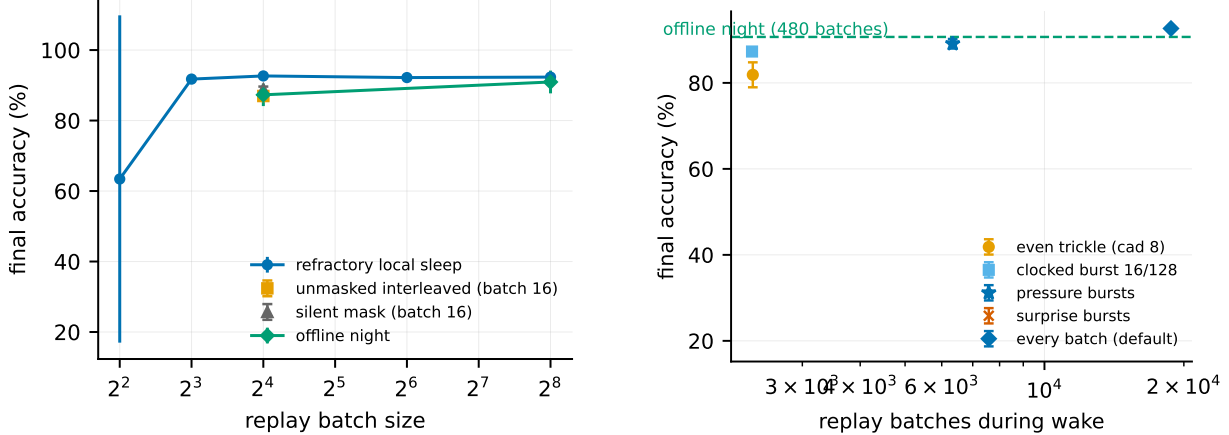


Figure 3: Left (isolation): final split-MNIST accuracy vs. replay batch size (record substrate, stateless SGD). Masked refractory replay is insensitive down to 8 samples; unmasked, silent-masked and offline-night replay all degrade. **Right (timing):** accuracy vs. number of waking replay batches. Even trickles fail; clocked bursts recover most of the gap; pressure-triggered bursts (star) dominate at sparse budgets; surprise triggering never fires at all.

corrected cell shares its bookkeeping with every other row.) Second, replacing masked Adam with *stateless* heavy-ball SGD inside the same isolation raises the ceiling: $92.7 \pm 0.3\%$ sequential and $94.7 \pm 0.3\%$ static (six seeds; vs. $91.8 \pm 0.6/94.6 \pm 0.2$ for masked Adam; unmasked SGD control 87.0 ± 1.1). Carrying no state at all is simpler and better — and closer to biology; the ablation in Section 6 traces masked Adam’s deficit to the confinement of its moments, which the exactness guarantee demands but accuracy, it turns out, does not. Third, *soft* rotation (halving, rather than silencing, recent winners) loses where it matters (90.2 ± 0.4 sequential, 86.4 ± 0.3 static): all-or-none OFF periods beat turned-down gain, echoing the biological finding that tonic firing-rate reduction does not discharge local sleep pressure while ON/OFF alternation does (Driessen et al., 2026). Table 1 in Section 6 collects these controls together with the remaining component knockouts; a two-sided learning-rate sweep (Appendix E) shows the comparison is a plateau rather than a knife-edge, with masked Adam already at its own best.

5.4 When replay must be rare: homeostatic bursts, never surprise

Figure 3 (right): at 12.5% of replay events, a clocked burst of 16 batches every 128 retains 87.3 ± 0.2 while an even trickle at the same budget collapses (81.9 ± 2.9) — masked replay reaches only part of the synapses per event and cannot dribble. Unit-level pressure triggering does better still (89.1 ± 1.1 at a third of the events), and surprise triggering fails outright: at the record fraction the converged error never crosses the surprise threshold, so *zero* replay events fire (19.0, the forgetting floor). The waking trigger must be homeostatic — the opposite of the night’s optimal trigger, which our earlier controller showed *should* be novelty-timed. (The same ordering held throughout the historical 5% phases, where the surprise trigger fired occasionally and still failed at every volume.)

5.5 The i.i.d. price of rotation

Rotation’s i.i.d. price depends on the activity fraction. At the historical 5% it cost 1.3–3.0 points on i.i.d. MNIST and *reversed* into a +2.5 to +6.5-point static gain on split CIFAR-10 (forced rotation as a regulariser in the underfit regime); at the 10% record fraction the MNIST price shrinks to

≈ 1.2 points (94.7 vs. 95.9 rotation-free) and the CIFAR static effect becomes statistically neutral (27.6 ± 0.4 vs. 29.5 ± 2.8). Under masked Adam, a relative-novelty/progress gate (Eqs. (7)–(8)) removed the 5% price with one setting across all four regimes; the full Adam-era gate study — including why absolute thresholds provably cannot serve and why unit-level pressure gating only trades the axes — is in Appendix B. Under the stateless optimiser the gate had to be committed to bouts (Section 6); at the record fraction both cures become largely unnecessary — the sparsity dial (Section 6.3) shrinks the price at source.

5.6 Transfer to CIFAR-10

On split CIFAR-10 (grayscale, same protocol) the ordering transfers and widens: refractory local sleep $29.5 \pm 0.8 >$ offline night $26.2 \pm 0.6 >$ BP+ER $25.1 \pm 1.5 >$ unmasked ER $17.0 \pm 6.1 \approx$ no buffer 16.4, despite task-active code overlap of 0.9–0.98. On the V1-like feature front-end the local ordering persists (refractory $41.5 \pm 1.1 >$ offline night $35.0 \pm 0.5 \gg$ unmasked $20.1 \pm 7.3 >$ no buffer 17.3) and BP+ER reaches 43.2 ± 1.0 at its best learning rate — the reverse of the raw-pixel ordering. In our first report (at the historical 5% fraction, local learner at 36.8) this read as a 4.7-point regime boundary; a two-sided fairness pass plus the sparsity dial revise it to 1.7 (41.5 ± 1.1 local vs. 43.2 ± 1.0 BP+ER): most of the boundary was the local learner’s optimiser and its starved activity fraction, not its learning rule. On raw pixels the local learner leads outright.

The residual gap then decomposes entirely into the *substrate*, not the learning rule. Transplanting the local network’s ingredients into the BP net one at a time (each cell at its better of two learning rates): its width *helps* BP+ER (44.5 ± 1.0 for a dense 512–256 net), its 30% distance-dependent wiring — drawn by the same generator on the same sheets — costs 1.2 (43.3 ± 0.9), and its k -WTA costs the rest (40.5 ± 0.7 at the record 10%; 39.3 ± 1.0 at the historical 5%), leaving backprop-with-replay *at or below* the local learner on the identical substrate (41.5 ± 1.1 vs. 40.5 ± 0.7 at 10%; all cells six seeds except the local one, three). What separates the two systems on feature inputs is dense activation, not backpropagation’s credit assignment: on the substrate this project commits to, the local rule gives nothing away. The advantage over *other local schedules* (nights, unmasked ER) remains robust across all three input regimes.

5.7 Energy

Converted to spiking inference ($T_s=8$; firing thresholds calibrated on training images), the default system (always-on rotation, stateless SGD, record fraction; rate accuracy 94.7% on the measured seed) runs at 93.8% and 119 nJ/sample vs. 2461 nJ for dense rate inference ($20.7\times$) and 526 nJ for event-driven rate inference ($4.4\times$), improving further with the time budget (94.4% at $T_s=16$, 241 nJ). The calibration interaction we reported for the earlier Adam stack has *reversed sign* under the stateless optimiser: the rotation-trained networks are now the calibration-robust ones (accuracy monotone in T_s), while the rotation-free control degrades beyond $T_s=8$ ($94.6 \rightarrow 90.1$ at $T_s=32$) — consistent with rotation shaping sparser codes (final hidden dimensionality 12.7 vs. 28.8; 67 vs. 82 spikes per sample at $T_s=8$).

5.8 Negative results

For the record, the following were falsified under matched budgets: a frozen dentate-gyrus-style sparse expansion front-end (input decorrelation does not propagate through learned k -WTA layers: task overlap at the input drops $0.79 \rightarrow 0.18$ yet every downstream metric worsens by 3–11 points); generative (REM) replay as a substitute for episodic samples; margin-based reverse learning; surprise-gated memory *writing*; multiplicative burst coding; absolute-threshold rotation gates;

Table 1: Component ablation of the consolidation mechanism on the record substrate (512–256/10%; backprop references dense 256–128). Sequential = final accuracy on split-MNIST; static = i.i.d. MNIST (the axis that must not degrade). Dashes: combination not applicable. Bold: column best. Headline rows carry six seeds; the rest three.

Variant	Sequential	i.i.d. (static)
<i>Full system and single knockouts (stateless SGD, $\eta = 0.02$)</i>		
full: exact isolation + rotation + per-batch progress gate	89.7 ± 1.4	95.3 ± 0.2
the same gate committed to bouts (min on-duration 2048)	92.3 ± 0.2	94.9 ± 0.3
– gate (rotation always on; <i>the default</i>)	92.7 ± 0.3	94.7 ± 0.3
+ two-timescale anchor ($\lambda=\mu=3\cdot 10^{-4}$), rotation always on	92.7 ± 0.4	94.6 ± 0.3
– rotation (pre-silent isolation only)	88.8 ± 0.8	95.9 ± 0.2
– isolation (unmasked interleaved replay)	87.0 ± 1.1	96.1 ± 0.1
– replay (no episodic buffer; Adam)	19.6 ± 0.0	95.8 ± 0.2
<i>Optimiser state under the mask (rotation always on, Adam)</i>		
moments confined to the mask	91.8 ± 0.6	94.6 ± 0.2
moments advancing outside the mask	92.6 ± 0.3	94.3 ± 0.3
<i>Replacing the construction</i>		
offline night instead of concurrent replay, same substrate	90.9 ± 3.2	—
the same night on its preferred narrow substrate	90.7 ± 0.4	—
mirror direction: protect the past (Section 5.3)	77.9 ± 1.6	92.3 ± 0.2
soft rotation: gain 0.5 instead of 0	90.2 ± 0.4	86.4 ± 0.3
<i>External references</i>		
unmasked ER at the same budget (Adam)	89.7 ± 1.1	95.4 ± 0.3
backprop (+ER sequential; plain static), dense	89.3 ± 0.8	97.5 ± 0.0

pruning anchored to sparse replay bursts (Appendix D); soft (gain-reduction) rotation and mirror-direction isolation (Section 5.3); an OFF-side gate refractory — the flip-flop’s inactive half in this system; strongly coupled synaptic anchors; and utility-exempt rotation as a route past the bout gate (all Section 6).

6 Ablation Study

The system under test is a conjunction of five ingredients: a biologically constrained substrate, the exact isolation mask (6), the refractory rotation, a stateless optimiser, and the progress-keyed gate (8). This section removes or replaces one ingredient of the consolidation mechanism at a time (Table 1). The substrate beneath it was fixed by earlier ablation ladders; its full cost accounting — 1.2 points below a dense backpropagation MLP at $4\times$ fewer synaptic operations per pass — is reproduced in Appendix A. Every mechanism cell shares the record substrate (512–256 hidden, 10% k -WTA, 30% distance-dependent connectivity), the same buffer ($K=1000$, random writing) and the same replay budget (waking batch 16, replay batch 16 after every waking batch); the appendices retain the corresponding grids at the historical 5% fraction, where every ordering below was first established.

6.1 Reading the component table

Read top-down, the first block prices each ingredient by what its removal costs the sequential axis under the stateless optimiser: rotation is worth +3.9 points (92.7 vs. 88.8 for the pre-silent

channel alone), isolation +5.7 with a four-fold variance reduction (92.7 ± 0.3 vs. 87.0 ± 1.1), and replay is worth everything (19.6 is the forgetting floor). The static column identifies who pays: rotation is the only statically costly component (94.7 always-on against 95.9 – 96.1 without it), a price of ≈ 1.2 points at the record fraction. The gates that were built to refund a larger price at the historical 5% fraction are here *neutral*: the bout-committed gate trades -0.4 sequential for $+0.2$ static ($92.3 \pm 0.2/94.9 \pm 0.3$), the per-batch gate is erratic ($89.7 \pm 1.4/95.3 \pm 0.2$), and the two-timescale anchor no longer adds anything ($92.7 \pm 0.4/94.6 \pm 0.3$). Their mechanisms remain instructive — at 5% the per-batch gate’s flicker destroyed the stateless gain, episode commitment (the sleep–wake flip-flop’s consolidated-bout structure (Saper et al., 2005)) repaired it, and bout-length and OFF-refractory controls located the residual price in the rotation of settled coalitions itself — and that full study is preserved in Appendix F and the 5% grids; but at the record fraction the dial has absorbed most of what they bought, and the default system is simply always-on rotation.

The middle block tests the exactness requirement itself and retains the table’s sharpest surprise. Propositions 1–2 require Adam’s moments to advance only inside the mask: state advancing outside re-injects the replay gradient into awake synapses through the next unmasked waking step, and output-neutrality is lost. Empirically, however, the leak *helps* sequentially — 92.6 ± 0.3 vs. 91.8 ± 0.6 . Exact state confinement, not isolation, is what holds masked Adam below the stateless optimiser: moments frozen outside each batch’s mask go stale across a unit’s on/off cycles (Bricken et al., 2023), and either remedy — letting the state track every gradient (surrendering the guarantee) or carrying no state at all — recovers most of the gap. Statelessness is the only choice that keeps the guarantee and the accuracy at once.

One budget note completes the block: half the cadence with doubled batches costs 1.3 (91.4 ± 0.3) — the stateless system prefers consolidation to run continuously.

6.2 Two probes of the residual static price

At the historical 5% fraction, the OFF-refractory null located the bout gate’s residual static price in the rotation of settled coalitions itself. Two mechanisms attacked that diagnosis directly (details in Appendix F; 5% numbers). *Utility-exempt rotation* (sparing the top- q long-use units from ever being benched) only traces a trade-off that the bout gate dominates: temporal commitment beats structural exemption. *Two-timescale synapses* in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016) fail to dissolve the price, but weak symmetric coupling acts as a *sequential stabiliser* at 5% (92.4 ± 0.2 , variance halved, static parity); at the record fraction the anchor is neutral ($92.7 \pm 0.4/94.6 \pm 0.3$). Anchor and bout gate compose sub-additively. Figure 4 draws the operating frontier.

6.3 The sparsity dial

The substrate’s original 5% k -WTA fraction was fixed in an early phase and inherited by a year of experiments; the decomposition just given made it suspect — if sparse k -WTA taxes BP by points on features, it presumably taxes our own learner too. It did, on both regimes, and the tax was bought back almost for free; this subsection documents the dial that led to the 10% record fraction used throughout the main text. On feature CIFAR-10, relaxing the fraction improves *both* axes monotonically: sequential $40.5 \rightarrow 42.7 \pm 0.3$ and static $39.4 \rightarrow 45.6 \pm 0.5$ from 5 to 25%, with forgetting slightly *reduced*; the BP control traces the same curve ($39.3 \rightarrow 40.9 \rightarrow 43.3$), with the local learner 1.0–1.3 points ahead at every matched level. On MNIST the sequential optimum sits at 10%: always-on rotation reaches **92.7 ± 0.3** sequential at 94.7 ± 0.3 static (six seeds) — dominating the entire historical 5% family, bout gate and anchor included — while 15% trades to

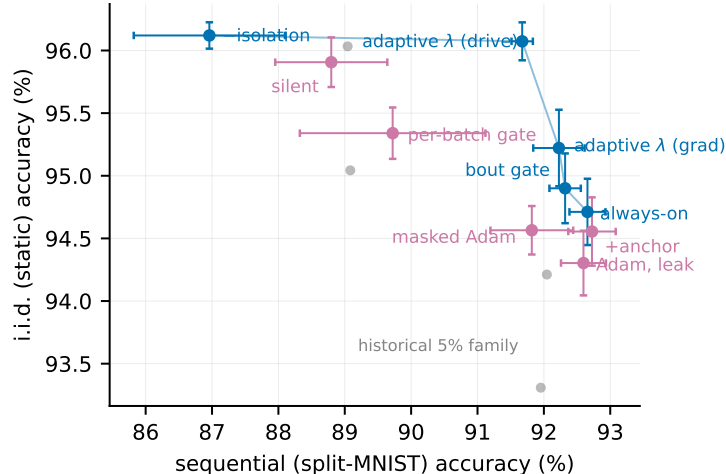


Figure 4: The operating frontier at the record fraction under the stateless optimiser: split-MNIST sequential accuracy against i.i.d. static accuracy (headline points six seeds). Blue: non-dominated configurations; purple: dominated ones; small grey dots: the historical 5% family the project inherited, entirely inside the new frontier. The two adaptive- λ points (Section 7) extend the static side of the frontier and retire the silent mask and the per-batch gate. No single configuration wins both axes; always-on rotation is the sequential default.

$92.5 \pm 0.3/95.1 \pm 0.3$ and 25% plateaus ($92.5 \pm 0.1/95.1 \pm 0.2$).

Three controls sharpen the reading. First, the no-buffer ceiling itself rises with the fraction ($95.1 \rightarrow 95.8 \rightarrow 95.9$), so rotation’s static price *at matched sparsity* shrinks from 1.8 to 0.7–1.0 points — and the gate’s *raison d’être* shrinks with it: at 15%, always-on rotation no longer trails the bout-gated system. Second, rotation-free silent isolation gains nothing from the dial ($89.0 \rightarrow 89.3 \rightarrow 88.7$): the dial helps specifically the *rotating* family, which at 5% was starved of representational room. Third, asymmetric schedules (a mild input-facing layer above a sparse deep layer) fail — sequential parity with uniform relaxation but a 3–4-point static deficit: the tax is levied at every layer. The dial is free elsewhere: synaptic operations rise only 3–5%, and spiking inference stays input-dominated (93.7% at 120 nJ, $T_s=8$, at 15%; monotone in T_s as before). Following this finding the record substrate moved to 10%: Table 1 and every main-text number now use it, with the complete mechanism grid re-established there (orderings preserved throughout), while the appendices retain the historical 5% grids on which each mechanism was first isolated. The dial, not the gate, is the cheapest cure for rotation’s static price (Figure 4).

7 An adaptive decay controller

The last tuned constant. Every result above runs the rule of (5) with a decay λ tuned once per dataset (10^{-3} on the image benchmarks). Split CIFAR-100 — ten tasks of ten classes — exposes what that constant hides. With one hundred classes, each class receives a tenth of the per-epoch supervision events, while the decay acts on every synapse at every step, undiminished; at the MNIST-tuned value the hidden-to-hidden weights follow the pure-decay trajectory $(1 - \lambda)^t$ *exactly* (the thinned error signal loses the race outright), deep activity collapses by $5\times$, the readout gradient goes to zero, and the network lands at chance — 2.4% i.i.d. against 20–23% for a ten-class subset of the same images on the same substrate. The class count of the supervision signal is a hidden variable that every fixed λ is silently tuned against. A manual rescue ($\lambda=10^{-4}$, $3\times$ targets) revives

Table 2: One dimensionless constant replaces per-dataset tuning. Sequential / static accuracy (%) under the tuned fixed decay (10^{-3} on the image benchmarks; 10^{-4} with $3\times$ targets as the CIFAR-100 rescue) against the controller of (9) at the same $\rho = 0.25$ everywhere. Gradient-referenced: six seeds; drive-referenced: three; the full system (refractory rotation, isolated replay, $K=1000$) in every cell.

	fixed λ (tuned)		ctrl. (gradient)		ctrl. (drive)	
	seq.	static	seq.	static	seq.	static
split-MNIST	92.7 ± 0.3	94.7 ± 0.3	92.2 ± 0.4	95.2 ± 0.3	91.7 ± 0.2	96.1 ± 0.2
split CIFAR-10	29.5 ± 0.8	27.6 ± 0.4	28.1 ± 0.9	31.2 ± 0.8	26.5 ± 1.3	29.9 ± 0.3
feature CIFAR-10	41.5 ± 1.1	41.8 ± 1.1	40.5 ± 0.6	43.8 ± 0.2	38.9 ± 2.0	43.0 ± 1.2
split CIFAR-100	1.4 ± 0.5	1.1 ± 0.1	3.7 ± 0.4	4.3 ± 0.2	3.4 ± 0.6	3.7 ± 0.3
+ features	1.2 ± 0.3	1.2 ± 0.2	5.1 ± 0.8	—	5.7 ± 0.8	6.0 ± 0.7

activity but survives the full protocol only at 1.1–1.4%.

The controller. The project’s design principle — control signals must be self-referenced, never absolute (Section 4.3) — applies at the synapse. We replace the constant by a per-layer ratio controller,

$$\lambda_\ell = \min\left(\lambda_{\max}, \rho \frac{\text{EMA}[\overline{|\Delta W^\ell|}]}{\overline{|W^\ell|}}\right), \quad \rho = 0.25, \quad \lambda_{\max} = 0.02, \quad (9)$$

where $\overline{|\Delta W^\ell|}$ is the mean magnitude of the update the optimiser actually applied to layer ℓ and the EMA runs over waking, unmasked steps only; W^ℓ and $B^{\ell-1}$ share λ_ℓ , so Kolen–Pollack alignment is untouched. The decay removes a fixed fraction of what learning currently adds: layers and regimes with rich supervision keep a strong regulariser, starved ones are automatically spared. A gradient-referenced variant ($\rho \eta \text{EMA}[\overline{|g^\ell|}]/\overline{|W^\ell|}$) behaves equivalently; we report both, with the drive-referenced form canonical because it is optimiser-agnostic.

One constant, every dataset. Table 2 gives the cross-dataset verdict, with a consistent signature. On the sequential axis the controller concedes 0.5–1.4 points to the per-dataset tuned value; on every static axis it wins, by +0.5 to +3.6 — the fixed decay had under-regularised every thick-signal regime all along — and split-MNIST static 96.1 ± 0.2 is the best i.i.d. accuracy the two-hidden-layer substrate has produced under *any* configuration, isolation-free included. On CIFAR-100 the controller lifts the system $3\text{--}5\times$ over the manual rescue with no dataset-specific knob at all (floor-regime numbers, reported for ordering only: the strong BP+ER reference reaches 6–10% there). The realised λ_ℓ (logged throughout) self-differentiates exactly as the tuning history would predict — $\sim 10^{-5}$ on MNIST sequential streams, $\sim 10^{-4}$ on static ones, up to 3×10^{-3} on raw CIFAR, layer-graded within each net — so the per-dataset λ ladder was a signal-to-decay ratio in disguise. On the operating frontier (Figure 4) the two controller points are non-dominated and retire the silent mask and the per-batch novelty gate; the tuned fixed decay remains the sequential-axis record holder.

Depth is the same boundary, and two mechanisms cross it. The class count thins the error signal per class; depth attenuates it per layer — the same starvation on a different axis. A depth ladder (three, four and five hidden layers, 512–256–128..., everything else the record configuration) makes the parallel exact: under the tuned fixed decay the static axis dies at four hidden layers ($9.7 \pm 0.7\%$, chance) and the sequential axis turns bimodal at five (55.5 ± 39.3), while under the controller every cell stays alive and degrades gracefully — $90.6 \rightarrow 90.0 \rightarrow 87.6$ sequential, $95.6 \rightarrow 95.3 \rightarrow 91.8$ static — with the realised λ_ℓ dropping to 3×10^{-6} in the starved middle

layers. The residual toll is then architectural, and a biologically unexceptional bypass removes it. ResNet-style skip synapses — S^ℓ carrying $a^{\ell-2}$ into each deep hidden layer, learned by the same local product with a Kolen–Pollack feedback twin and the shared per-layer decay, and replay-masked under the same pre-or-post-asleep rule — restore the thick error path, and the five-hidden-layer system returns to the two-layer record on *both* axes: 91.5 ± 0.4 sequential, 96.2 ± 0.3 static, with 96.3 ± 0.2 at depth four the best i.i.d. accuracy of the project. The bypass alone also rescues the fixed decay ($91.3/93.6$ at depth five), locating the depth collapse in error-path attenuation rather than decay mis-tuning per se; the controller still adds its usual static margin on top of the bypass. A BP+ER reference at the same widths is depth-insensitive (≈ 88 – 89 sequential throughout) but forgets 2 – $3\times$ more ($F \approx 13$ – 14 vs. 5 – 7); the local system with skips and the controller beats it at every depth tested (three seeds). The two starvations are not interchangeable, however: on split CIFAR-100 depth is a net loss that skips only half refund (four hidden layers $1.9/1.6 \rightarrow 3.8/3.0$ with skips, against $5.7/6.0$ at two), whereas *width* at the short chain length lifts the local record to $6.7 \pm 0.3/7.5 \pm 0.2$ (1024 – 512 ; a $5\times$ buffer adds nothing) — the thin-signal regime wants capacity, not depth, and BP+ER’s remaining lead there (9.9 ± 0.5) comes with twice the forgetting.

What the controller must never see, and what it does not do. Composition follows one rule: the drive estimate must be fed only *unamplified waking updates*. Inflated one-hot targets and the offline night’s $3\times$ -gain batches both push the EMA up until λ_ℓ hits its clip and erases day learning — night $\times$ controller collapses to 60.2 ± 2.8 against 90.9 ± 3.2 for the night alone — whereas mechanisms that respect the flag compose cleanly: anchor and bout gate under the controller keep the standard signature (bout + controller reaches 96.0 ± 0.3 static). And the controller deliberately does not subsume λ ’s second role: on small tabular problems the decay earns its keep as a *capacity* regulariser against overfitting (cf. Appendix A), information that no drive ratio contains; where generalisation, not alignment, sets the correct decay, a tuned constant should keep the job.

8 Discussion

Why isolation stabilises micro-batches. A replay micro-batch is a high-variance gradient estimate. Applied to the full network it perturbs the very coalition currently encoding the wake stream, and with shared optimiser state the perturbation compounds (cf. the unmasked curve of Figure 3). Inside the mask, each update leaves the present input’s hidden computation untouched — exactly for silent-presynaptic and suppressed units, and for all but $\sim 0.3\%$ of waking samples per step in measurement (Propositions 1–2 and the Remark) — so variance accumulates only where it can do no immediate harm and is averaged over thousands of events. The offline night enjoys no such protection: with nothing clamping the network, a small noisy batch at $3\times$ learning rate moves the whole weight state.

Correspondence with biology. We treat the biology as motivating analogy, not as a claim of mechanistic equivalence. The refractory rule mirrors the causal finding of [Driessen et al. \(2026\)](#) that ON/OFF *alternation*, not tonic silencing, discharges local sleep pressure and rescues memory — our silent-mask control (their tonic condition) is reliably worse. The dissociation of triggers — homeostatic for waking consolidation, novelty-timed for full nights — is a testable prediction for biology. The regime dependence of the rotation price (cost on easy streams, gain under chronic underfitting) may bear on why local sleep in vivo both impairs behaviour ([Vyazovskiy et al., 2011](#)) and serves useful functions when appropriately timed.

Honest limits. Every configuration choice across the project’s phases was made on the official test sets — no validation split existed — so the absolute numbers carry an unquantified selection margin; Appendix G replicates the headline orderings on a held-out tenth of the training data never trained on (Spearman $\rho = 0.92$, mean gap 0.9 points), and those numbers are the honest estimates. All results are at MLP scale on MNIST/CIFAR variants with three seeds (six on the headline rows); BP+ER keeps a 1.7-point lead on feature-based CIFAR with both sides at their best — a lead the substrate decomposition attributes to dense activation rather than to backpropagation itself; rotation keeps a ≈ 1.2 -point static price at the record fraction that no gate we tested removes without a sequential cost (Section 6); the same-substrate offline night is wildly unstable (90.9 ± 3.2), a phenomenon we report but do not explain; and the novelty claims rest on our own literature search (August 2026); individual primitives — refractory competition, sparse or null-space isolation, replay, adaptive scheduling — all have ancestors, and the claimed contribution is their combination into concurrent, exactly isolated consolidation.

9 Future Work

(i) The bimodal instability of the same-substrate offline night at the record fraction (90.9 ± 3.2) — what makes offline consolidation fragile on wide substrates; (ii) the bout gate’s residual static price (94.2 vs. 96.0 rotation-free): the OFF-side refractory, utility exemption and synaptic anchoring are now all ruled out as routes past it (Section 6), leaving mastery-annealed rotation depth untried; (iii) the narrow-vs-wide substrate interaction of offline nights at tiny buffers; (iv) convolutional stacks — depth and residual-style routing are answered (Section 7: five hidden layers at the two-layer record with locally learned skips and the controller; on 100-class streams width, not depth, is what pays), convolution and a learned front-end are not, and the static-axis wall of the 100-class regime (6–7.5% under every substrate change) says the front-end is the binding constraint there; (v) a throughput theory of the isolated channel predicting the required replay cadence from ρ_ℓ , code overlap and the interference rate; (vi) the optimiser-dependence of the rotation–spiking calibration interaction, whose sign reverses between the Adam and the stateless stack; (vii) a drive estimate that survives amplified update streams (inflated targets, night gains), which currently defeat the controller’s clip.

10 Conclusion

Consolidation does not require an offline phase. With exactly isolated replay, refractory rotation of the active coalition, homeostatic burst triggering and a relative-novelty gate on the rotation itself, a biologically constrained learner consolidates *while it infers*: it matches and slightly exceeds the strongest offline-night schedule on split-MNIST at $\sim 1.2\times$ the night’s replay samples, pays nothing on learnable i.i.d. streams, gains outright on harder data, and runs at a twentieth of dense-rate inference energy when spiking. The offline night retains one irreducible role — consolidating very small episodic stores — which sharpens, rather than blurs, the division of labour between wake and sleep.

References

- Abbasi, A., Nooralinejad, P., Braverman, V., Pirsiavash, H., Kolouri, S. (2022). Sparsity and heterogeneous dropout for continual learning in the null space of neural activations. *CoLLAs*, PMLR 199.
- Akrout, M., Wilson, C., Humphreys, P., Lillicrap, T., Tweed, D. (2019). Deep learning without weight transport. *NeurIPS* 32.

- Arani, E., Sarfraz, F., Zonooz, B. (2022). Learning fast, learning slow: A general continual learning method based on complementary learning system. *ICLR*.
- Bazhenov, A., Delanois, J.E., Krishnan, G.P. (2026). Not just after one: Sleep-inspired replay prevents catastrophic forgetting after sequential tasks. *arXiv:2606.08447*.
- Borbély, A.A. (1982). A two process model of sleep regulation. *Human Neurobiology* 1, 195–204.
- Benna, M.K., Fusi, S. (2016). Computational principles of synaptic memory consolidation. *Nature Neuroscience* 19, 1697–1706.
- Bricken, T., Davies, X., Singh, D., Krotov, D., Kreiman, G. (2023). Sparse distributed memory is a continual learner. *ICLR*.
- Cayco-Gajic, N.A., Clopath, C., Silver, R.A. (2017). Sparse synaptic connectivity is required for decorrelation and pattern separation in feedforward networks. *Nature Communications* 8, 1116.
- Cayco-Gajic, N.A., Silver, R.A. (2019). Re-evaluating circuit mechanisms underlying pattern separation. *Neuron* 101, 584–602.
- Driessen, K., Squarcio, F., Tononi, G., Cirelli, C. (2026). Induction of cortical ON/OFF periods in awake mice fulfills sleep functions. *Nature Neuroscience* 29, 1954–1965.
- Farajtabar, M., Azizan, N., Mott, A., Li, A. (2020). Orthogonal gradient descent for continual learning. *AISTATS*, PMLR 108.
- Golden, R., Delanois, J.E., Sanda, P., Bazhenov, M. (2022). Sleep prevents catastrophic forgetting in spiking neural networks by forming a joint synaptic weight representation. *PLoS Computational Biology* 18, e1010628.
- Golkar, S., Kagan, M., Cho, K. (2019). Continual learning via neural pruning. *arXiv:1903.04476*.
- Harun, M.Y., Gallardo, J., Hayes, T.L., Kemker, R., Kanan, C. (2023). SIESTA: Efficient online continual learning with sleep. *TMLR*.
- Hasselmo, M.E. (2006). The role of acetylcholine in learning and memory. *Current Opinion in Neurobiology* 16, 710–715.
- Iyer, A., Grewal, K., Velu, A., Souza, L.O., Forest, J., Ahmad, S. (2022). Avoiding catastrophe: Active dendrites enable multi-task learning in dynamic environments. *Frontiers in Neurobotics* 16.
- Kapoor, K., Mackey, W., Aloimonos, Y., Lin, X. (2026). HiCL: Hippocampal-inspired continual learning. *AAAI*.
- Kaski, S., Kohonen, T. (1994). Winner-take-all networks for physiological models of competitive learning. *Neural Networks* 7, 973–984.
- Klasson, M., Kjellström, H., Zhang, C. (2023). Learn the time to learn: Replay scheduling in continual learning. *TMLR*.
- Kolen, J.F., Pollack, J.B. (1994). Backpropagation without weight transport. *IEEE ICNN*.
- Kubo, Y., Delanois, J.E., Bazhenov, M. (2025). Toward lifelong learning in equilibrium propagation: Sleep-like and awake rehearsal for enhanced stability. *arXiv:2508.14081*.
- Lässig, F., Aceituno, P.V., Sorbaro, M., Grewe, B.F. (2023). Bio-inspired, task-free continual learning through activity regularization. *Biological Cybernetics* 117, 345–361.
- Litwin-Kumar, A., Harris, K.D., Axel, R., Sompolinsky, H., Abbott, L.F. (2017). Optimal degrees of synaptic connectivity. *Neuron* 93, 1153–1164.

- Maeda, M., Miyajima, H. (1999). Competitive learning methods with refractory and creative approaches. *IEICE Trans. Fundamentals* E82-A(9), 1825–1833.
- Mallya, A., Lazebnik, S. (2018). PackNet: Adding multiple tasks to a single network by iterative pruning. *CVPR*.
- Masse, N.Y., Grant, G.D., Freedman, D.J. (2018). Alleviating catastrophic forgetting using context-dependent gating and synaptic stabilization. *PNAS* 115, E10467–E10475.
- McKee, K., Hazy, T., Zheng, Y., Bugaud, Z., Miconi, T. (2026). Cortex-inspired continual learning: Unsupervised instantiation and recovery of functional task networks. *arXiv:2604.24637*.
- Merolla, P.A., et al. (2014). A million spiking-neuron integrated circuit with a scalable communication network and interface. *Science* 345, 668–673.
- Pham, Q., Liu, C., Hoi, S. (2021). DualNet: Continual learning, fast and slow. *NeurIPS* 34.
- Robinson, B.S., Lau, C.W., New, A., Nichols, S.M., Johnson, E.C., Wolmetz, M., Coon, W.G. (2022). Continual learning benefits from multiple sleep mechanisms: NREM, REM, and synaptic downscaling. *arXiv:2209.05245*.
- Ross, G.J., Adams, N.M., Tasoulis, D.K., Hand, D.J. (2012). Exponentially weighted moving average charts for detecting concept drift. *Pattern Recognition Letters* 33, 191–198.
- Saha, G., Garg, I., Roy, K. (2021). Gradient projection memory for continual learning. *ICLR*.
- Saper, C.B., Scammell, T.E., Lu, J. (2005). Hypothalamic regulation of sleep and circadian rhythms. *Nature* 437, 1257–1263.
- Shen, Y., Dasgupta, S., Navlakha, S. (2021). Algorithmic insights on continual learning from fruit flies. *arXiv:2107.07617*.
- Shen, J., Ni, W., Xu, Q., Tang, H. (2024). Efficient spiking neural networks with sparse selective activation for continual learning. *AAAI* 38, 611–619.
- Sorrenti, A., Bellitto, G., Proietto Salanitri, F., Pennisi, M., Palazzo, S., Spampinato, C. (2024). Wake-sleep consolidated learning. *arXiv:2401.08623*.
- Tadros, T., Krishnan, G.P., Ramyaa, R., Bazhenov, M. (2022). Sleep-like unsupervised replay reduces catastrophic forgetting in artificial neural networks. *Nature Communications* 13, 7742.
- Tilley, M.J., Miller, M., Freedman, D.J. (2023). Artificial neuronal ensembles with learned context dependent gating. *ICLR*.
- Tononi, G., Cirelli, C. (2014). Sleep and the price of plasticity: From synaptic and cellular homeostasis to memory consolidation and integration. *Neuron* 81, 12–34.
- Vyazovskiy, V.V., Olcese, U., Hanlon, E.C., Nir, Y., Cirelli, C., Tononi, G. (2011). Local sleep in awake rats. *Nature* 472, 443–447.
- Wang, S., Li, X., Sun, J., Xu, Z. (2021). Training networks in null space of feature covariance for continual learning. *CVPR*.
- Whittington, J.C.R., Bogacz, R. (2017). An approximation of the error backpropagation algorithm in a predictive coding network with local Hebbian synaptic plasticity. *Neural Computation* 29, 1229–1262.

Table 3: Substrate ablation on i.i.d. MNIST (256–128 hidden core, three seeds). “Syn. ops” is the fraction of the dense MLP’s synaptic operations in one forward pass. Upper block: constraints added cumulatively; lower block: one ingredient replaced at its stage.

Substrate	Accuracy	Syn. ops
dense backpropagation MLP (autograd reference)	97.7 ± 0.1	1.00
local burst-error learner, unconstrained	97.8 ± 0.1	0.85
+ bounded error, Dale’s law, sign-concordant feedback (no transport)	97.1 ± 0.0	0.85
+ k -WTA 10%	97.0 ± 0.2	0.80
+ relaxation $T=5$	97.0 ± 0.1	0.80
+ 30% distance-dependent connectivity	96.5 ± 0.2	0.24
+ burst gate and burst baseline	96.3 ± 0.2	0.24
+ single-phase burst sweep replacing relaxation (<i>final substrate</i>)	96.5 ± 0.1	0.24
random wiring instead of distance-dependent, equal density	95.4 ± 0.0	0.24
one settling step ($T=1$) instead of five	17.9 ± 6.8	0.80
weight transport restored (mirroring, no KP decay)	96.3 ± 0.2	0.24
Kolen–Pollack decay removed	96.3 ± 0.1	0.24
heavy-ball SGD instead of Adam ($\eta = 0.02$)	95.2 ± 0.3	0.24
k -WTA 5% / 2% instead of 10%	95.9 ± 0.1 / 94.2 ± 0.2	0.79/0.78
multiplicative burst coding	78.8 ± 1.9	0.24
training on $T_s=8$ spike counts	80.0 ± 0.8	0.24

A The price of the substrate

Table 3 traces the path from a dense backpropagation MLP to the substrate used throughout, on i.i.d. MNIST where each constraint’s cost is cleanest. The upper block is cumulative; the lower block replaces one ingredient at its stage and is to be compared with the matching cumulative row.

The cumulative ladder prices the entire biological constraint set at 1.2 points below the dense backpropagation reference ($97.7 \rightarrow 96.5$) for $4\times$ fewer synaptic operations per pass; the local burst-error objective itself costs nothing (97.8 unconstrained). No single constraint is expensive: the bounded-error/Dale/no-transport trio costs 0.7, 10% k -WTA one tenth of a point, and the 70% connectivity cut half a point — of which distance-dependent wiring recovers $+1.1$ over random wiring at equal density. The lower block adds three cautions. Settling cannot merely be shortened ($T=1$ collapses to 17.9); the single-phase burst sweep is the principled replacement, matching the $T=5$ accuracy in a single pass. The Kolen–Pollack decay and even fully restored weight transport are accuracy-neutral at this scale — the decay earns its keep as a regulariser on harder inputs, not as an aligner. And the two spiking-flavoured training schemes fail outright: spiking is affordable at *inference* (the energy measurements above), not as a training-time code.

Finally, the two tables disagree about the optimiser in an instructive way: on the plain substrate SGD *loses* 1.3 points to Adam (95.2 vs. 96.5), yet inside the isolated-consolidation loop it wins on both axes. Per-synapse adaptive state is an asset while every synapse trains on every batch and a liability once updates become intermittently mask-confined — the staleness mechanism of Table 1’s middle block seen from the other side. The optimiser is not a neutral implementation detail: it interacts with the sleep machinery, and the right choice reverses between the static substrate and the consolidating system.

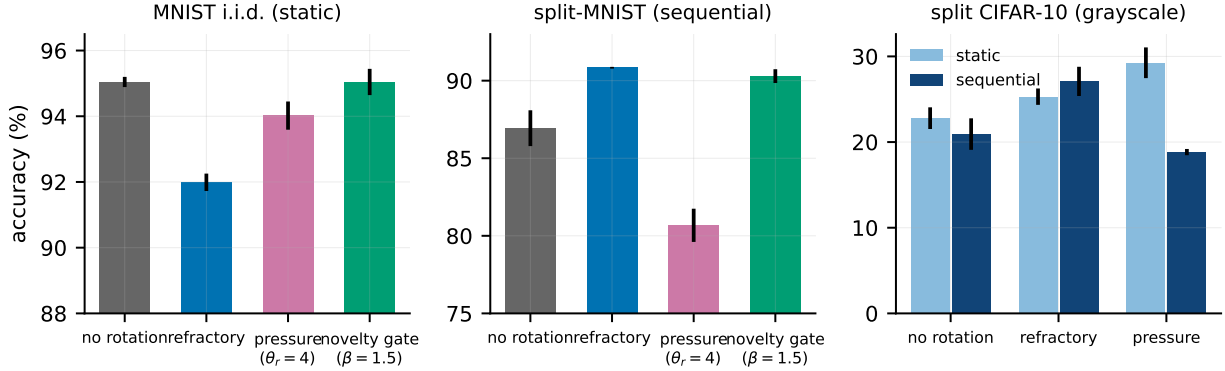


Figure 5: Rotation: price, trade-off, and gate. Left: on i.i.d. MNIST the novelty gate restores no-rotation accuracy. Middle: on split-MNIST rotation is required and the gate keeps it. Right: on split CIFAR-10 (grayscale) rotation is a gain on *both* axes.

B The per-batch rotation gates under masked Adam

Figure 5: always-on rotation costs 1.3–3.0 points on i.i.d. MNIST ($95.0 \rightarrow 92.0$ – 93.7). Gating rotation by *unit-level pressure* only trades the axes (static $91.6 \rightarrow 94.0$ as sequential $88.8 \rightarrow 80.7$; no dominant point). The *relative novelty* gate (7) removes the price at no sequential cost on MNIST: static 95.0 ± 0.4 (rotation on 1.4–17% of batches), sequential 90.3 ± 0.4 ($\beta=1.5$) or 90.9 ± 0.6 ($\beta=1.1$). On split CIFAR-10 the price *reverses*: rotation gains +2.5 (refractory) to +6.5 points (pressure-selected rest) on the static axis — forced rotation acts as a regulariser in the underfit regime — while the pure novelty gate (7) *misfires* there (sequential 19.1 ± 1.1 : fast/slow $\rightarrow 1$ on a stream that never becomes predictable, so rotation shuts off exactly where it helps). The progress-keyed gate (8) resolves this with one setting across all four regimes: split CIFAR-10 sequential 27.2 ± 0.7 and static 25.3 ± 1.0 (both = always-on rotation), split-MNIST 90.4 ± 0.9 , i.i.d. MNIST 94.3 ± 0.2 at $\gamma=0.5$ — retaining a residual 0.7-point static cost relative to the pure novelty gate on easy streams, the price of the unmastered clause. (All gate numbers in this subsection use masked Adam; under the stateless optimiser the gate must be committed to minimum-duration bouts — Section 6.)

C Small buffers: what the night still buys

At $K=200$ the night appeared to keep a ~ 2.6 -point edge in our earlier phases; the final triangulation (Figure 6) dissolves most of it into a substrate mismatch. On the *same* 512–256/5% substrate, the night reaches 84.0 ± 3.4 while noisy pressure-burst local sleep reaches 84.3 ± 1.0 and local+night 84.8 ± 0.9 — parity, with three-fold smaller variance for the wake-side scheme. The strongest tiny-buffer number remains the narrower 256–128/10% night (86.9), so the offline window retains an advantage only in combination with narrow substrates. The mechanism results stand: replay diversity (noise $\sigma=0.2$) is specifically a wake-side lever — it lifts every local scheme ($81.8 \pm 2.9 \rightarrow 84.3 \pm 1.0$) and *hurts* the night ($84.0 \rightarrow 81.8 \pm 3.2$) — and *unmasked* waking replay collapses (71.0 ± 1.7): isolation is a benefit, not a cost, at every buffer size we tested.

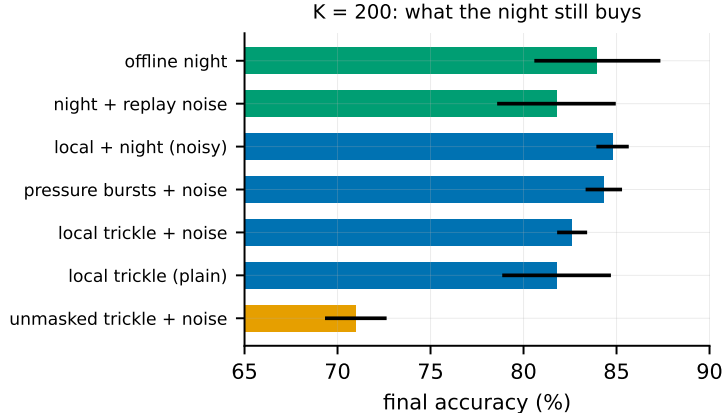


Figure 6: $K = 200$ mechanism triangulation. Isolation helps, diversity helps only during wake, and the irreducible advantage of the night at tiny buffers is its interference-free window.

D Sleep-anchored down-selection

Motivated by the synaptic homeostasis hypothesis (Tononi and Cirelli, 2014), we anchored structural plasticity (prune the weakest fraction of live synapses, regrow the same number with a distance bias) to consolidation windows, using the Kolen–Pollack decay of (5) as the natural weakening force on unreplayed synapses. The verdict splits three ways at matched turnover ($\approx 5\%$ per epoch). On the *continuous* local-sleep substrate pruning is free (sleep-anchored 90.9 ± 0.2 , clocked 90.6 ± 0.4 , none 90.8 ± 0.1). On the pure *night* substrate, pruning immediately after each night helps substantially (87.2 ± 3.8 vs. 84.1 ± 1.3) — offline down-selection rescues the wide-sparse night. On the *episodic-burst* substrate pruning is harmful (84.6 ± 0.5 burst-anchored, 86.6 ± 1.3 clocked, vs. 89.9 ± 0.1 without): when replay is rare, synapses are cut before enough consolidation events have protected them. Down-selection, like replay itself, needs either density or an offline window.

E Learning-rate sensitivity

The stateless comparison invites an objection: SGD’s η was chosen, Adam’s was left at its default. A two-sided sweep dissolves it. Under always-on rotation, SGD holds a sequential plateau over $\eta \in [0.01, 0.02]$ ($91.9 \pm 0.5/92.0 \pm 0.6$) and a static plateau over $[0.02, 0.05]$ ($93.3\text{--}93.7$), decaying gracefully outside ($86.3/89.2$ at $\eta=0.1$); $\eta=0.02$ is the joint optimum. Masked Adam, swept over a tenfold range, has its sequential optimum exactly at the default (90.8 ± 0.1 at 10^{-3} ; 88.6 at $3 \cdot 10^{-4}$, 88.9 at $3 \cdot 10^{-3}$) — the comparison of Section 5.3 was already at Adam’s best — and no Adam setting reaches the stateless joint point ($3 \cdot 10^{-3}$ buys 93.6 static at a three-point sequential cost). Unmasked SGD is poor and unstable at *every* rate ($84\text{--}88$, seed deviations $1.8\text{--}4.1$, three- to eight-fold those of the masked runs), making the stabilising role of isolation a cross- η fact. The bout-committed gate inherits the plateau, and the optimum shifts one notch down on CIFAR ($\eta=0.01$).

F Probes of the residual static price

Bout length and the OFF-side refractory. Bout length is a proper knob: $M=512$ (duties $0.61/0.38$) gives $91.7 \pm 0.4/94.0 \pm 0.5$, already recovering most of what the per-batch gate loses, while $M=4096$ drives the static duty to 0.94 and the refund vanishes (93.4 ± 0.1). The static ceiling

of silent isolation (96.0) stays out of reach — rotation, even in bouts, keeps a residual price — so the committed gate buys regime-adaptivity, not a free lunch. Completing the flip-flop with an OFF-side refractory (once a bout expires, triggers are ignored for M_{off} batches) is a clean null: short refractories ($M_{\text{off}} \leq 1024$) never engage on the static stream — a bout expires only after its trigger has been quiet for M batches, so static re-triggering is not flicker to begin with — while a long one ($M_{\text{off}} = 4096$) lowers the static duty to 0.53 without moving static accuracy (94.0 ± 0.7) and monotonically erodes the sequential axis ($92.1 \rightarrow 91.0$, occasionally masking a task switch). Together with the $M=512$ cell (duty 0.38, static 94.0), this locates the residual static price in the rotation of settled coalitions itself, not in how often the gate fires.

Structural exemption and synaptic anchoring. *Utility-exempt rotation* spares the top- q units by long-term use trace from ever being benched, so the settled coalitions stay awake. It traces a smooth trade-off between always-on rotation and the rotation-free system ($q=0.10$: $90.9 \pm 0.1/94.2 \pm 0.2$; $q=0.25$: $89.1 \pm 0.4/95.4 \pm 0.2$), but the bout gate beats the $q=0.10$ point by +1.1 sequential at equal static: *temporal commitment dominates structural exemption*, recapitulating at the utility level what unit-level pressure gating showed earlier (Figure 5) — any structural narrowing of the rotation pool is paid for in replay-channel width.

Two-timescale synapses give every weight a slow anchor in the spirit of synaptic consolidation cascades (Benna and Fusi, 2016): the fast weight decays toward its anchor at rate λ while the anchor absorbs it at rate μ , ticking only on waking updates so the exactness of the replay path is untouched. The strong hypothesis — that settled function living in the anchor would dissolve rotation’s static price — is falsified: across $\lambda \in [3 \cdot 10^{-4}, 3 \cdot 10^{-3}]$, $\mu \in [10^{-4}, 10^{-3}]$ no cell exceeds the anchor-free static accuracy, and strong coupling hurts both axes (the anchor becomes a lagging drag). Weak symmetric coupling ($\lambda=\mu=3 \cdot 10^{-4}$), however, acts as a *sequential stabiliser*: 92.4 ± 0.2 , the best sequential cell of the 5% grid with the seed variance halved, at static parity — weakly dominating the 5% always-on point. Anchor and bout gate compose sub-additively ($92.3 \pm 0.3/93.8 \pm 0.5$, between the parents).

G Held-out replication of the headline orderings

Every configuration choice made during this project — some four hundred sequential-axis configurations over twenty phases — was made on the official test sets; no validation split existed. The absolute numbers in the main text are therefore optimistic by an unknown selection margin. As a check on the *orderings*, which carry the claims, we re-ran the headline configurations with a tenth of the training set held out (stratified, fixed split, never trained on) as the evaluation set and the remaining nine tenths as the stream, three seeds each (Table 4). The held-out numbers sit 0.9 points below the test numbers on average (a smaller training set and a distribution-matched evaluation set both contribute), the sequential-axis ranking correlates at Spearman $\rho = 0.92$, and every claim the paper rests on survives: always-on rotation and the bout gate lead ($91.6/91.7$ vs. the night’s 90.7 ± 2.2 , BP+ER’s 88.7 and unmasked ER’s 83.7); the silent mask trails rotation; the five-hidden-layer skip system holds the two-layer level on both axes ($90.9/96.0$); and the adaptive decay keeps its static-side signature (95.3 vs. 94.0). The only inversions are within noise (rotation vs. bout gate, 0.1 points; the night vs. the controller, inside the night’s seed variance).

Table 4: Held-out replication. Sequential / static accuracy (%) on the official test set (as in the main text) and on a held-out tenth of the training set never used for training; seeds 0–2 in both columns, so the test column may differ from six-seed main-text entries.

	sequential		static	
	test	held-out	test	held-out
always-on rotation (default)	92.7 ± 0.3	91.6 ± 0.4	94.8 ± 0.3	94.0 ± 0.3
bout-committed gate	92.3 ± 0.3	91.7 ± 0.3	—	—
adaptive λ (drive)	91.7 ± 0.2	90.5 ± 0.7	96.1 ± 0.2	95.3 ± 0.0
five hidden layers + skips + adaptive λ	91.5 ± 0.4	90.9 ± 0.1	96.2 ± 0.3	96.0 ± 0.2
silent mask (no rotation)	89.4 ± 0.3	88.0 ± 0.6	—	—
offline night, same substrate	89.8 ± 4.7	90.7 ± 2.2	—	—
unmasked interleaved ER	87.0 ± 1.1	83.7 ± 3.8	—	—
BP + ER	89.3 ± 0.8	88.7 ± 0.4	—	—
no buffer	19.6 ± 0.0	19.4 ± 0.0	—	—